

TSEBO LAMULA

Graduate Geologist | Geophysicist | Hydrogeologist

Mamelodi, Pretoria, Gauteng | 079 927 5304 | tsebolamula7@gmail.com
LinkedIn: linkedin.com/in/tsebo-lamula-a9b421249 | SAGA Recognised Student Member

PROFESSIONAL PROFILE

Geology graduate with a completed Postgraduate Diploma in Geology and hands-on experience in multi-method geophysical surveys, hydrogeological investigations, mine water characterisation, and underground geological mapping. Demonstrated capability in ERT, gravity, magnetic and seismic refraction data acquisition and interpretation; hydrogeochemical sampling and PHREEQC modelling; salt and fluorescent dye tracer testing; and technical report writing. Seeking a Junior Geologist, Geophysicist, Hydrogeologist or Research Assistant role where geoscientific methods are applied to mining, environmental assessment and subsurface investigation.

PROFESSIONAL EXPERIENCE

Research Assistant – Hydrogeology & Mine Water

Mar 2025 – Apr 2026

Tshwane University of Technology (SARChI Chair for Mine Water Management) | Arcadia, Pretoria

- Supported hydrogeological and hydrogeochemical investigations at the abandoned Havelock Asbestos Mine (eSwatini), contributing to multi-tracer hydrodynamic studies of a flooded underground mine system ($\approx 8\text{--}11\text{ Mm}^3$ mine water volume).
- Executed underground and surface water sampling, salt tracer flow measurements, fluorescein tracer tests, tracer recovery monitoring and automatic water sampling to characterise mine water quality and hydraulic connectivity.
- Operated field instrumentation, captured and managed datasets in Microsoft Excel, and sorted and digitised geological plans and historical mine records of the Havelock Mine.
- Assisted with tracer injection logistics for NaCl, uranine and eosin-Y tests that quantified first arrivals (1.0–1.5 days), mean residence times (8–9 days) and flow velocities (0.6–1.3 m/min) over 1.1–1.4 km flow paths.

Mineralogy Tutor

Jul 2025 – Nov 2025

Tshwane University of Technology | Arcadia, Pretoria

- Delivered Mineralogy tutorials and practical sessions for undergraduate geology students, covering mineral identification, optical properties and laboratory techniques.
- Prepared mineral specimens, practical materials and laboratory equipment; designed mineral identification exercises; marked practical assessments and provided constructive feedback.
- Provided academic consultation and revision classes, supporting student performance in tests and examinations.

Mineralogy Tutor

Jul 2024 – Nov 2024

Tshwane University of Technology | Arcadia, Pretoria

- Conducted tutorial classes and academic support for undergraduate students in the Mineralogy module.
- Organised laboratory materials (mineral specimens, reagents, equipment); developed study materials; assessed practical and tutorial assignments and provided feedback.

Mining Junior Geologist

Sep 2022 – Sep 2023

Geostrat Training CC (South African Academy of Mining) | Randfontein

- Performed detailed underground face and development mapping, accurately recording lithology, structural features (faults, veins, alteration zones) and ore-zone boundaries under supervision of the site geologist.
- Collected and prepared underground chip, grab and channel samples in full compliance with company quality-control protocols and health-and-safety standards.
- Completed open-pit geological mapping and systematic sampling campaigns, collaborating with the supervising geologist and sampling team to ensure representative coverage.
- Compiled concise, data-driven geological reports after each underground and surface assignment, summarising key observations, assay results and recommendations for follow-up work.
- Documented critical underground geological structures contributing to improved resource modelling and grade-control planning.

EDUCATION

Postgraduate Diploma in Geology

Jan 2025 – Jun 2026

Tshwane University of Technology | Pretoria

Academic requirements completed June 2026 (official completion letter pending).

Relevant coursework: Geophysics; Mining & Exploration Geology; Engineering Geology; Hydrogeology.

Research Project: "A Multi-Method Geophysical, Hydrogeological and Hydrological Investigation of Post-Mining Environmental Effects at the Abandoned Lead Mine, Pretoria, South Africa."

- Designed and executed integrated geophysical surveys: 4 ERT lines, 5 gravity lines, 6 magnetic lines and 2 seismic refraction lines.
- Collected and analysed 16 water samples; performed PHREEQC speciation modelling and salt-tracer dilution flow measurements (downstream flow 594 L/min; mine contribution $\approx 290\text{ L/min}$).

- Identified low-resistivity anomalies (1.67–12.8 Ωm) correlating with shafts and slag heap; delineated fault-controlled mineralised zones and mining-induced subsidence features.

Advanced Diploma in Geology <i>Tshwane University of Technology Pretoria</i>	Jan 2024 – Dec 2024
Diploma in Geology <i>Tshwane University of Technology Pretoria</i>	Jan 2020 – Oct 2023
National Senior Certificate (Bachelor's Pass) <i>Lesedi Secondary School Donkerhoek, Bronkhorstspuit</i>	2015 – 2019

TECHNICAL SKILLS

Geophysics & Data Interpretation: ERT, Gravity, Magnetics, Seismic Refraction Tomography (SRT); data acquisition, processing and interpretation

Hydrogeology & Hydrogeochemistry: Water sampling, salt & fluorescent dye tracer testing, flow measurement, PHREEQC speciation modelling, mine water characterisation

Geological Field Methods: Underground face & development mapping, open-pit mapping, chip/grab/channel sampling, structural geology, ore-zone delineation, grade control support

Software & GIS: QGIS (spatial analysis & mapping), Microsoft Office Suite (Excel advanced data management, Word, PowerPoint), PHREEQC

Reporting & Research: Technical report writing, scientific research methods, data interpretation, laboratory mineral identification

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

Recognised Student Member – Southern African Geophysical Association (SAGA)	2026
Academic Orientation Programme for TUT Tutors – Tshwane University of Technology	Aug 2024
Certificate of Appreciation – Eskom Expo for Young Scientists	Aug 2024
Competency B Certificate – Mining Qualifications Authority (MQA)	Mar 2024

SOFT SKILLS & LANGUAGES

Soft Skills: Strong communication and interpersonal skills; problem-solving and critical thinking; time management and attention to detail; leadership and initiative in field and academic teams; adaptability to remote and underground site conditions.

Languages: English (Fluent), Sepedi (Fluent), IsiZulu (Fluent), Tshivenda (Fluent)

References available on request